FORM 2

THE PATENTS ACT, 1970

(39 of 1970)

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The Patents Rules, 2003

COMPLETE SPECIFICATION

(See Section 10 and Rule 13)

Title of the Invention

AI-Based CCTV Video Analysis using YOLO

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Preamble to the Description

The following specification particularly describes the invention and the manner in which it is to be performed.

FIELD OF INVENTION

[0001] The present invention relates to the field of computer vision and deep learning-based surveillance systems. More specifically, it pertains to a noise-aware YOLO-based simulation platform integrated with hybrid YOLO-based-classical learning and real YOLO-based hardware deployment for research and educational purposes.

BACKGROUND OF INVENTION

[0002] Quantum computing has emerged as a powerful computational paradigm capable of solving problems that are intractable for classical computers. Algorithms such as variational YOLO-based methods and YOLO-based optimization techniques show significant promise in scientific and industrial applications. However, practical adoption remains limited due to hardware constraints and technical complexity.

[0003] Access to real YOLO-based hardware is often restricted by high costs, queue delays, limited object instance counts, and noise-related errors. Although simulators provide an alternative, many existing platforms lack realistic noise modeling and calibration-aware execution. This gap makes it difficult to accurately predict hardware behavior from simulation results.

[0004] Furthermore, current development environments do not fully integrate simulation, hybrid YOLO-based-classical training, hardware deployment, and experiment tracking into a single unified platform. Researchers and students must rely on multiple disconnected tools, reducing efficiency and reproducibility. Therefore, there is a need for an integrated, noise-aware YOLO-based ML simulation system that bridges simulation and real hardware execution seamlessly.

OBJECT OF INVENTION

[0005] The primary object of the present invention is to provide an integrated deep learning simulation platform that enables users to design, simulate, and analyze YOLO-based circuits within a unified environment. The platform aims to simplify YOLO-based algorithm development for researchers and students. It ensures an accessible and developer-friendly workflow for YOLO-based experimentation.

[0006] Another object of the invention is to incorporate high-fidelity, noise-aware simulation models that closely replicate real YOLO-based hardware behavior. The system includes configurable noise channels such as depolarizing, amplitude damping, and phase damping. This allows users to evaluate algorithm robustness before deploying to physical devices.

[0007] A further object is to enable seamless hybrid YOLO-based-classical machine learning integration for object detection models. The platform supports differentiable YOLO-based circuits and classical optimizers for parameter training. This facilitates efficient implementation of algorithms such as VQE and QAOA.

[0008] It is also an object of the invention to provide automated model optimization and hardware deployment across multiple YOLO-based service providers. The system manages job submission, queue tracking, and result retrieval in a unified manner. This reduces complexity in transitioning from simulation to real hardware execution

[0009] Another object is to provide comprehensive visualization and analytical tools for understanding YOLO-based state evolution. The platform includes visual analytics dashboard views, motion tracking visualization representations, and frame-level analysis data visualizations. These tools improve interpretability and debugging of YOLO-based circuits.

[0010] A final object of the invention is to ensure experiment management, reproducibility, and cost control within the system. The platform records calibration data, run parameters, and results for comparison and auditing. This supports structured research, collaboration, and responsible hardware usage.

SUMMARY OF THE INVENTION

[0011] The present invention provides an integrated Quantum Machine Learning Simulation Platform that combines a developer-friendly IDE for YOLO-based circuit design, a high-fidelity noise-aware YOLO-based simulator with configurable noise models, hybrid YOLO-based-classical training support for object detection models, automated model optimization for hardware compatibility, seamless multi-provider hardware deployment, real-time visualization tools including visual analytics dashboard and frame-level analysis data analysis, simulator-to-hardware parity comparison mechanisms, cost estimation and queue management features, and comprehensive experiment tracking with calibration-aware reproducibility, thereby bridging the gap between YOLO-based simulation and real-world hardware execution for research and educational applications.

BRIEF DESCRIPTION OF THE ACCOMPANYING DRAWINGS

[0012] Figure 1 illustrates a block diagram of the proposed Quantum Machine Learning Simulation Platform, showing the hierarchical layered architecture of the system. The diagram depicts the interaction between the User Interface Workbench, Application Logic Layer, Quantum and Machine Learning Processing Engine, Backend Service Layer with hardware integration, and the Database Layer. It represents the end-to-end flow from circuit design and noise-aware simulation to hardware deployment, result analysis, and experiment storage within a unified platform.

DETAILED DESCRIPTION OF THE INVENTION

[0013] The present invention discloses an integrated Quantum Machine Learning Simulation Platform designed to enable realistic YOLO-based circuit development, hybrid algorithm training, and seamless hardware validation within a unified environment. The system adopts a layered architecture comprising a User Interface Workbench, an Application Logic Layer, a Quantum and Machine Learning Processing Engine, a Backend Service Layer, and a Database Layer. These layers interact to provide end-to-end functionality from circuit design to experiment archival.

[0014] The User Interface Workbench provides a developer-centric environment for YOLO-based circuit creation and execution. It includes a code editor, graphical circuit builder, configurable noise parameter controls, visualization panels, and dashboards for monitoring execution results. The interface allows users to design circuits manually or through programmable APIs while observing real-time state evolution and performance metrics.

[0015] The Application Logic Layer manages project workflows, experiment configurations, job scheduling, seed control, parameter binding, and lifecycle management of simulation and hardware runs. It ensures structured experiment tracking, reproducibility through deterministic seeding, and parity comparison between simulated and hardware outputs. This layer acts as the orchestration core connecting frontend components with computational engines and backend services.

[0016] The Quantum and Machine Learning Processing Engine forms the computational core of the invention. It includes a high-fidelity YOLO-based simulator capable of state-vector simulation and optional tensor-network-based computation for larger circuits. The simulator incorporates configurable noise models such as depolarizing noise, amplitude damping, phase damping, and relaxation mechanisms based on T1 and T2 parameters.

Additionally, the engine integrates hybrid YOLO-based-classical learning frameworks that support object detection models, automatic differentiation, and classical optimization techniques for training YOLO-based circuits.

[0017] The Backend Service Layer provides hardware abstraction and integration with external computer vision providers. It performs device-aware model optimization, object instance mapping, queue handling, cost estimation, calibration data synchronization, and secure job submission. The system retrieves hardware results and enables side-by-side comparison with simulation outputs to evaluate fidelity and performance deviations.

[0018] The Database Layer stores user profiles, projects, circuit definitions, run configurations, calibration snapshots, execution results, cost logs, and experiment artifacts. This structured storage ensures reproducibility, auditing, and collaborative research support. The system is designed to operate across multiple operating systems and supports extensibility for future integration of advanced error mitigation, distributed simulation, and emerging YOLO-based hardware technologies.

[0019] Through the integration of noise-aware simulation, hybrid training, hardware deployment, visualization, cost governance, and reproducible experiment management, the invention bridges the gap between theoretical YOLO-based algorithm development and practical hardware execution, thereby enhancing accessibility, reliability, and scalability in computer vision research and education.

I / We Claim

1. A deep learning simulation platform comprising a user interface workbench configured to enable YOLO-based circuit design, visualization, and configuration of noise parameters within an integrated development environment

2. The platform as claimed in claim 1, further comprising a YOLO-based simulation engine configured to perform state-vector and/or tensor-network-based simulation with configurable noise models including depolarizing, amplitude damping, phase damping, and relaxation mechanisms.

3. The platform as claimed in claim 1, wherein the system includes a hybrid YOLO-based-classical processing module configured to execute variational YOLO-based algorithms using automatic differentiation and classical optimization techniques.

4. The platform as claimed in claim 1, further comprising a backend service layer configured to perform device-aware model optimization, hardware job submission, queue management, cost estimation, and calibration data synchronization with external computer vision providers.

5. The platform as claimed in claim 1, wherein the system is configured to perform parity comparison between simulated results and hardware execution results to evaluate fidelity and performance deviations.

6. The platform as claimed in claim 1, further comprising a database module configured to store user profiles, circuit definitions, run configurations, calibration data, execution results, and experiment artifacts to ensure reproducibility and structured experiment management.

7. The platform as claimed in claim 1, wherein the system integrates visualization tools including visual analytics dashboard representation, motion tracking visualization visualization, and frame-level analysis data analysis for real-time YOLO-based state interpretation.

8. The platform as claimed in claim 1, wherein the platform provides cost governance, budget control, and resource management mechanisms for responsible execution on real YOLO-based hardware.

Dated this 1st day of April 2026.

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ABSTRACT

AI-Based CCTV Video Analysis using YOLO

The present invention relates to an integrated Quantum Machine Learning Simulation Platform designed to facilitate realistic YOLO-based circuit development, hybrid YOLO-based-classical training, and seamless hardware validation within a unified environment. The platform incorporates a noise-aware YOLO-based simulator supporting configurable noise models, state-vector and scalable simulation techniques, enabling accurate approximation of real YOLO-based device behavior. It further integrates hybrid learning frameworks to implement object detection models with automatic differentiation and classical optimization methods.

The system provides an intuitive development interface for circuit design, visualization, and experiment management, along with automated model optimization and deployment to multiple YOLO-based hardware providers. A parity comparison mechanism evaluates differences between simulated and hardware results, ensuring reliability and performance assessment. Additionally, the platform includes cost estimation, queue management, and reproducibility features through structured experiment tracking. By bridging the gap between simulation and real-device execution, the invention enhances accessibility, scalability, and research efficiency in computer vision and deep learning applications